

GPIE

Green Policy Intelligence Engine

All Maps and Plots

16 figures compiled from outputs/plots/

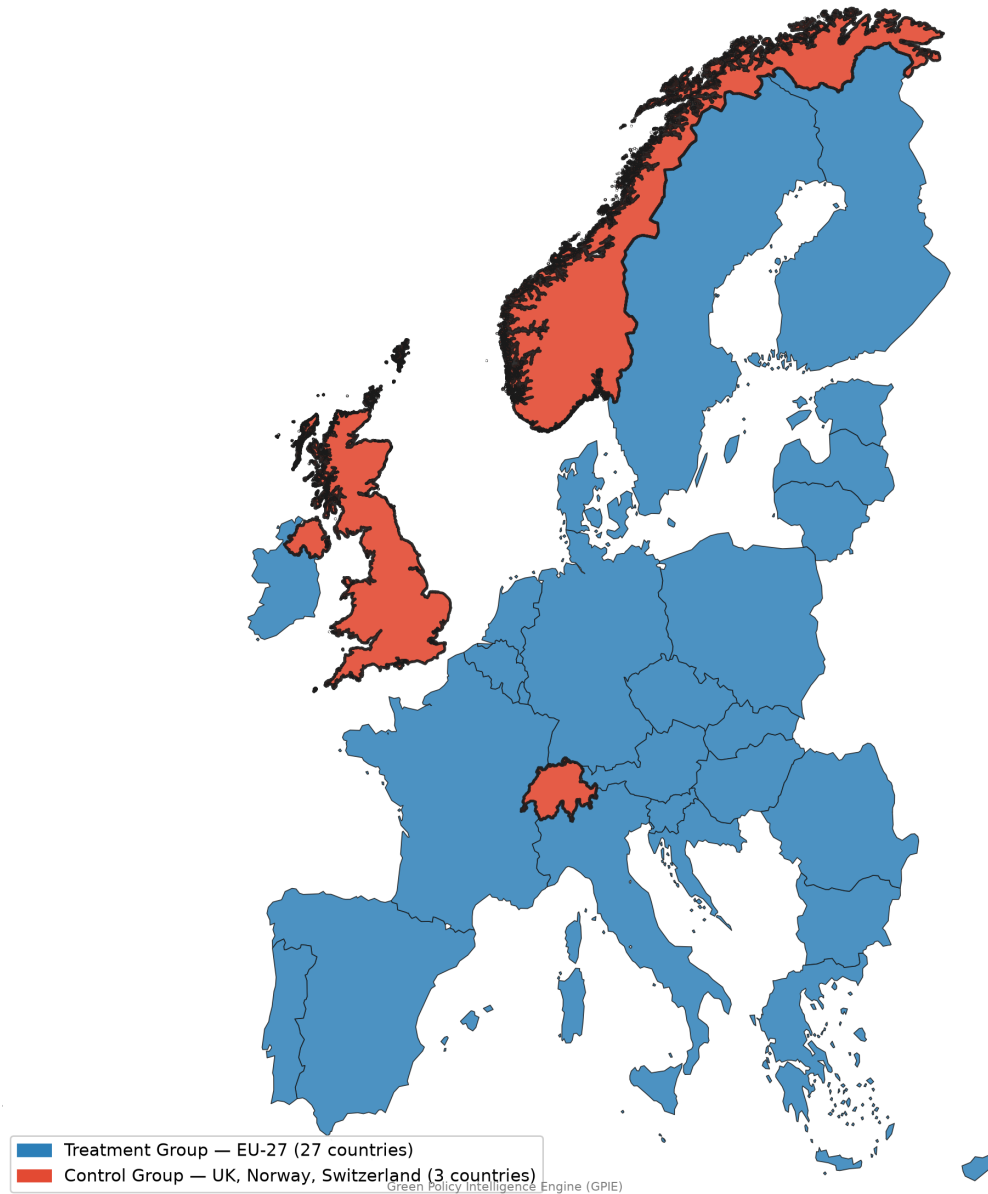
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3. NO2 Before vs. After the European Climate Law
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Study Design: Treatment vs. Control Group

EU-27 (treatment) and the non-EU control group - United Kingdom, Norway, Switzerland.

GPiE Study Design: Treatment vs. Control Group

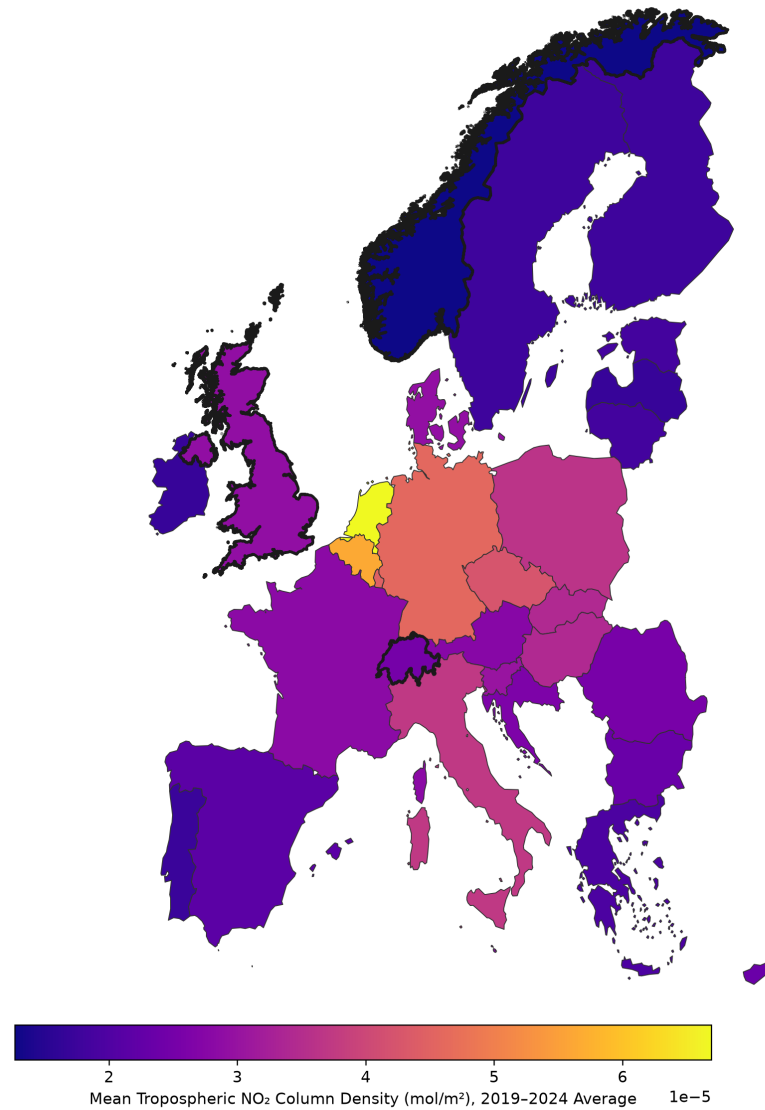
Difference-in-Differences comparison — EU-27 (subject to the European Climate Law) vs. three non-EU comparator countries



NO2 Choropleth Map

Mean tropospheric NO₂ concentration by country, Sentinel-5P TROPOMI.

Tropospheric NO₂ Concentration: EU-27 vs. Control Group (UK, Norway, Switzerland)
Thick borders mark non-EU control-group countries — visually comparable NO₂ levels support the DiD null result



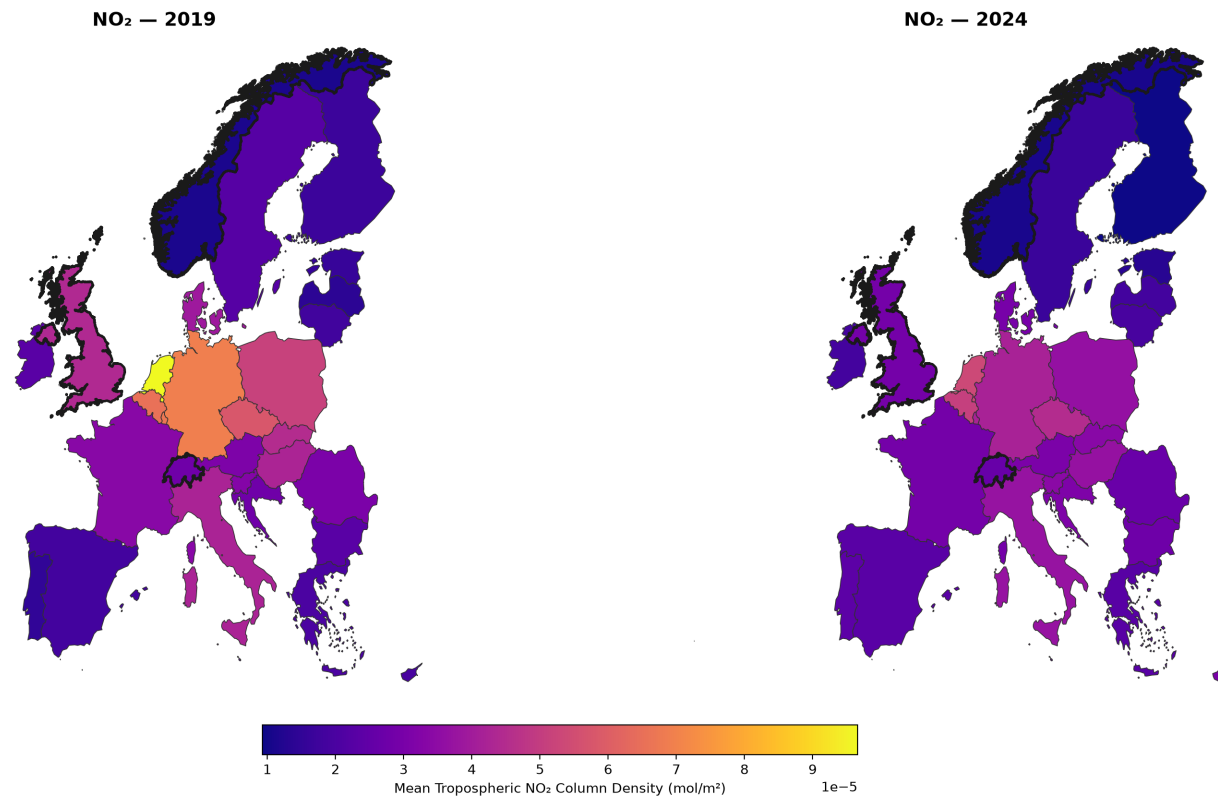
Green Policy Intelligence Engine (GPIE) — Source: Sentinel-5P TROPOMI, Sentinel Hub Statistical API

GPIE - Green Policy Intelligence Engine

NO2 Before vs. After the European Climate Law

Average NO2 across the EU-27, 2019 (pre-treatment) vs. 2024 (post-treatment).

NO₂ Levels Before vs. After the European Climate Law: EU-27 and Control Group (2019 vs. 2024)
Thick borders mark non-EU control-group countries (UK, Norway, Switzerland) — comparable decline pattern across both groups

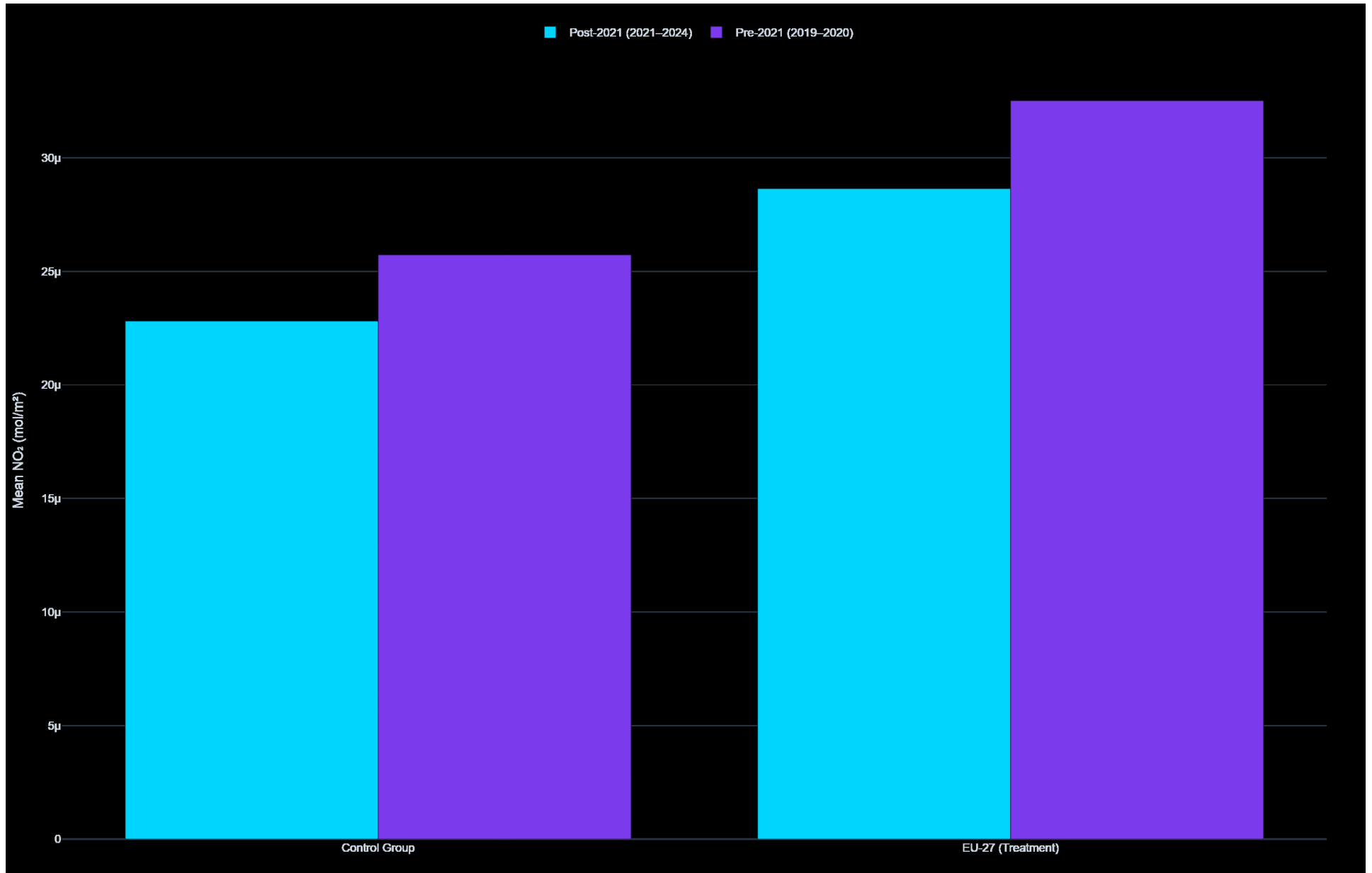


Green Policy Intelligence Engine (GPiE) — Source: Sentinel-5P TROPOMI, Sentinel Hub Statistical API

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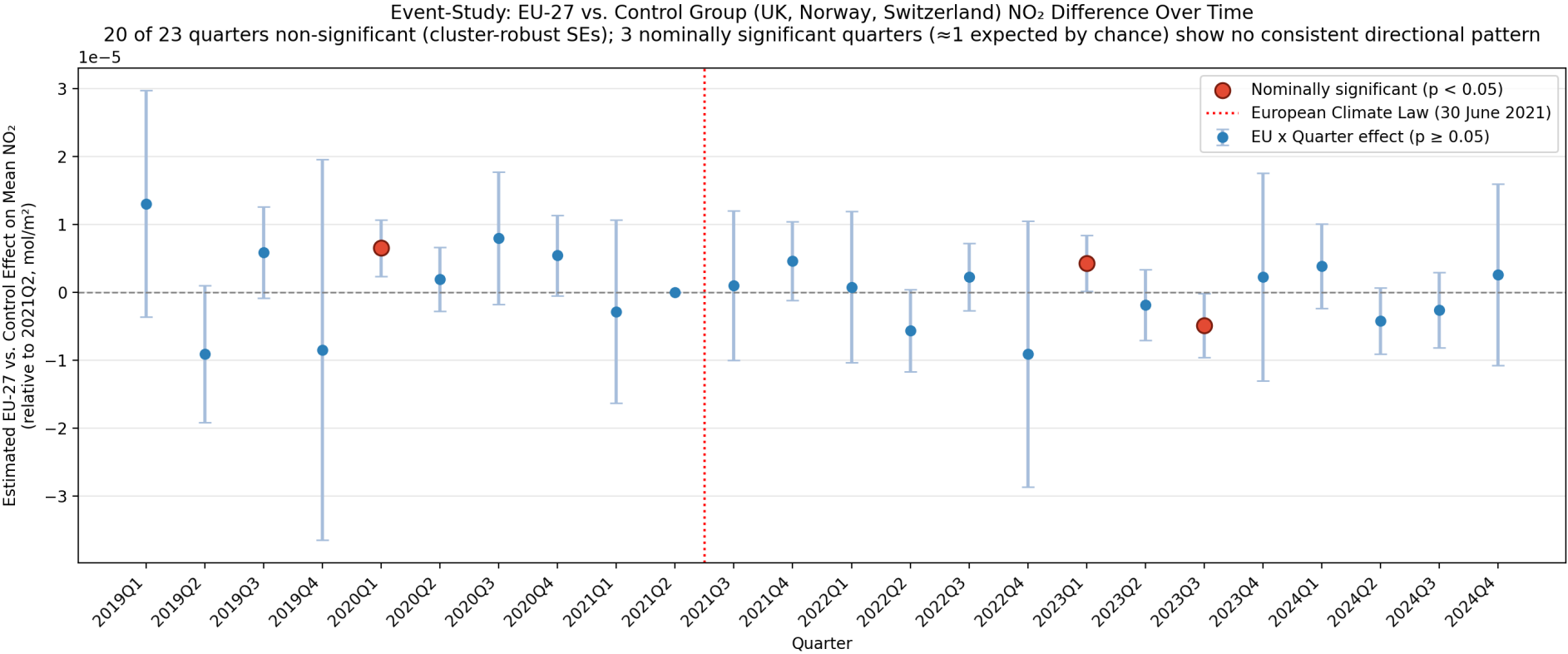
NO₂: EU-27 vs. Control Group

Mean NO₂, pre- and post-treatment, EU-27 vs. control group.



Event-Study: Quarter-by-Quarter NO2 Effect

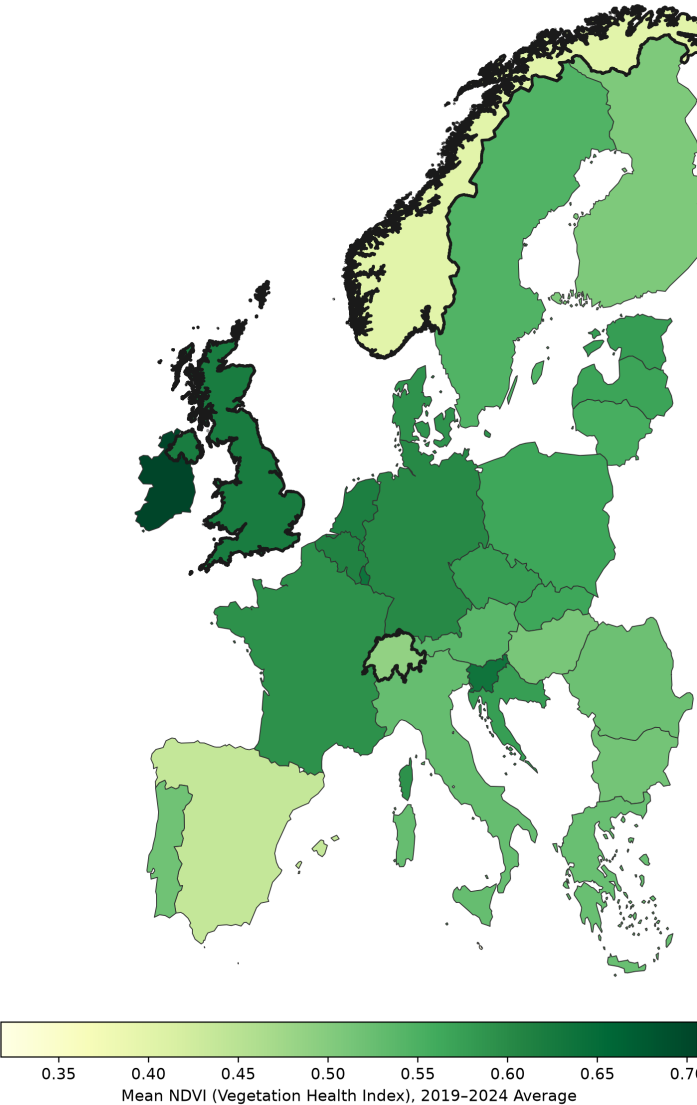
23-quarter Difference-in-Differences event-study estimates, cluster-robust SEs.



NDVI Choropleth Map

Mean vegetation health index (NDVI) by country.

Vegetation Health (NDVI): EU-27 vs. Control Group (UK, Norway, Switzerland)
Thick borders mark non-EU control-group countries



Green Policy Intelligence Engine (GPiE) — Source: CGLS NDVI 300m, Sentinel Hub Statistical API

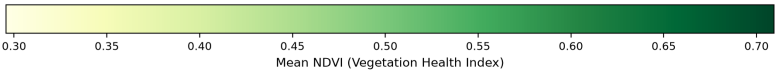
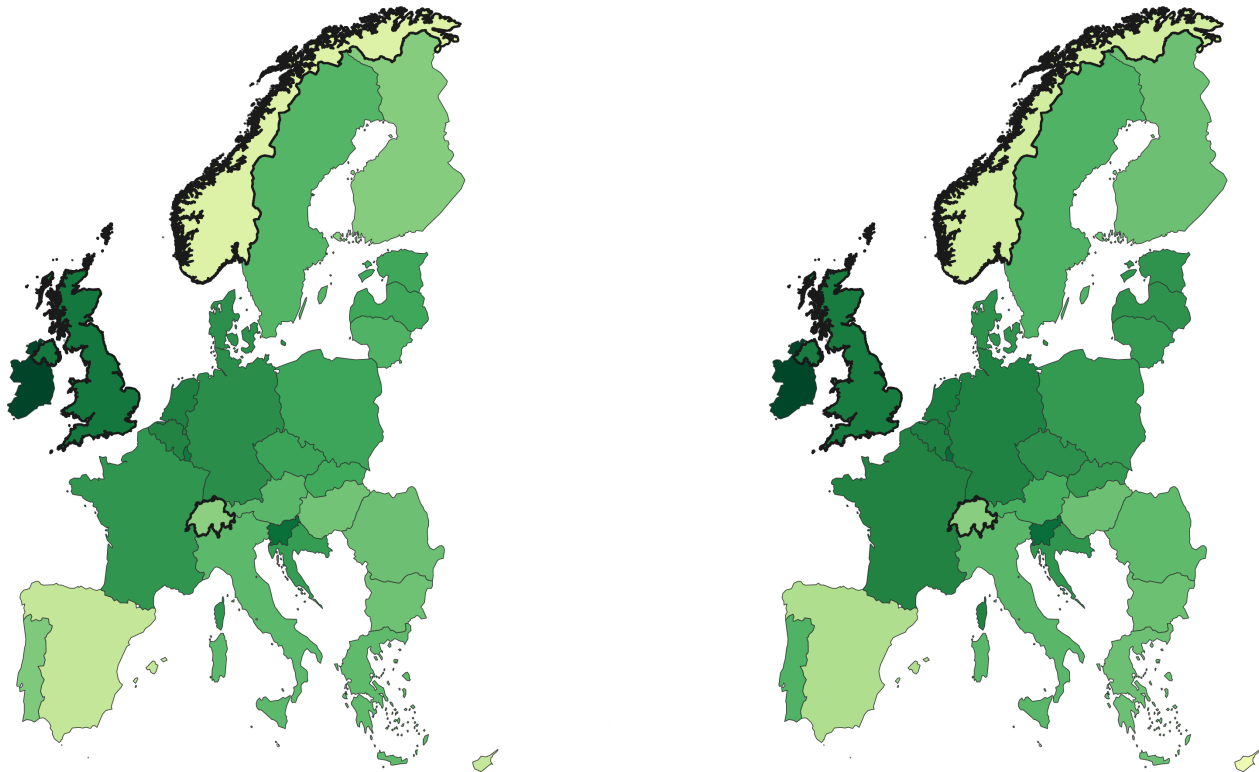
NDVI Before vs. After the European Climate Law

Average NDVI across the EU-27, pre- vs. post-treatment.

NDVI (Vegetation Health) Before vs. After the European Climate Law: EU-27 and Control Group (2019 vs. 2024)
Thick borders mark non-EU control-group countries (UK, Norway, Switzerland)

NDVI — 2019

NDVI — 2024

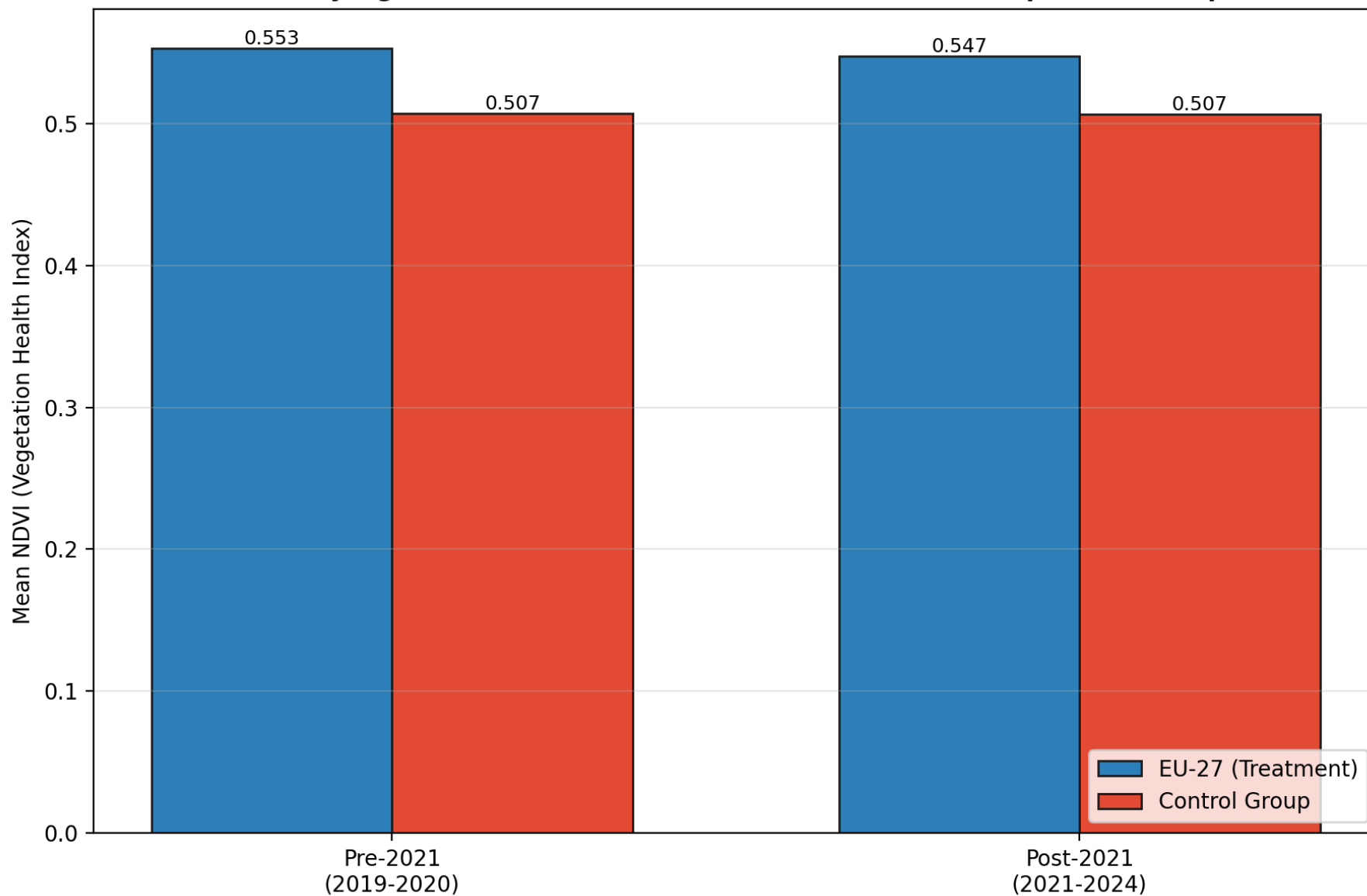


Green Policy Intelligence Engine (GPiE) — Source: CGLS NDVI 300m, Sentinel Hub Statistical API

NDVI: EU-27 vs. Control Group

Corrected two-group DiD model: coefficient = -0.021, p = 0.012 (cluster-robust).

NDVI: EU-27 vs. Control Group, Before vs. After the European Climate Law
Corrected two-group DiD model: coefficient = -0.021, p = 0.012 (cluster-robust) -
a statistically significant relative decline, not visible in NO2's equivalent comparison



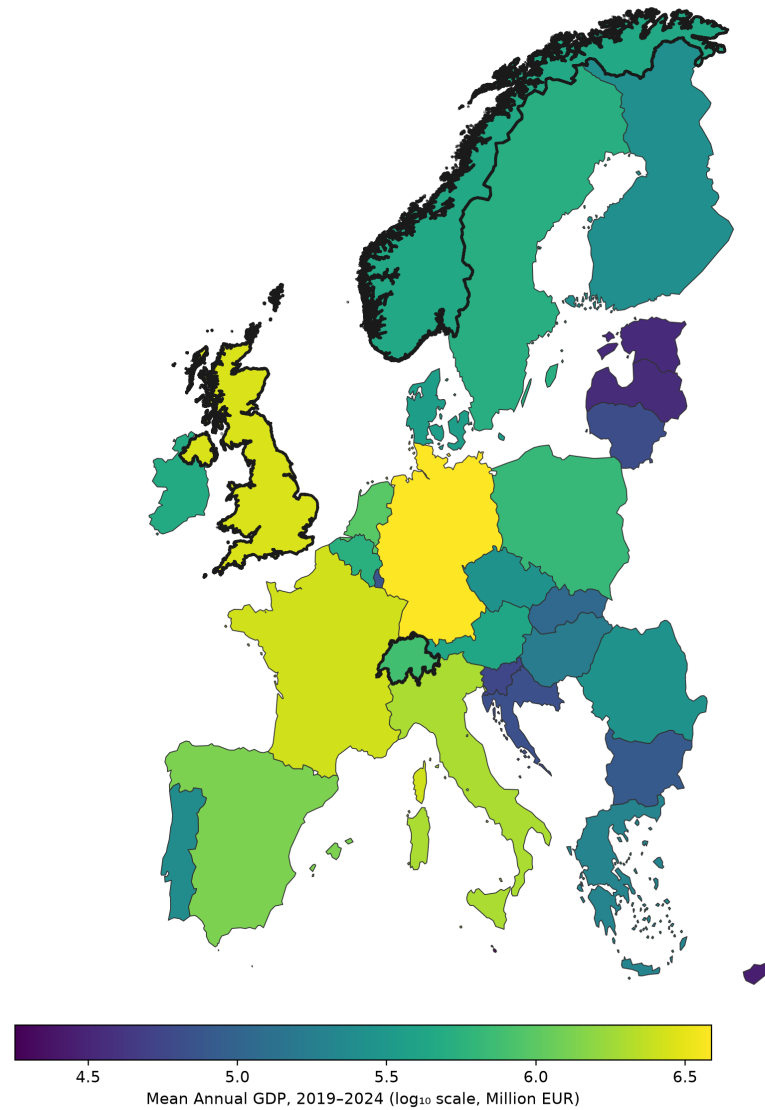
Green Policy Intelligence Engine (GPIE) - Source: CGLS NDVI 300m, Sentinel Hub Statistical API

GDP Choropleth Map

Control variable - GDP by country.

GDP: EU-27 vs. Control Group (UK, Norway, Switzerland)

Thick borders mark non-EU control-group countries — GDP used as an economic control variable in the causal model



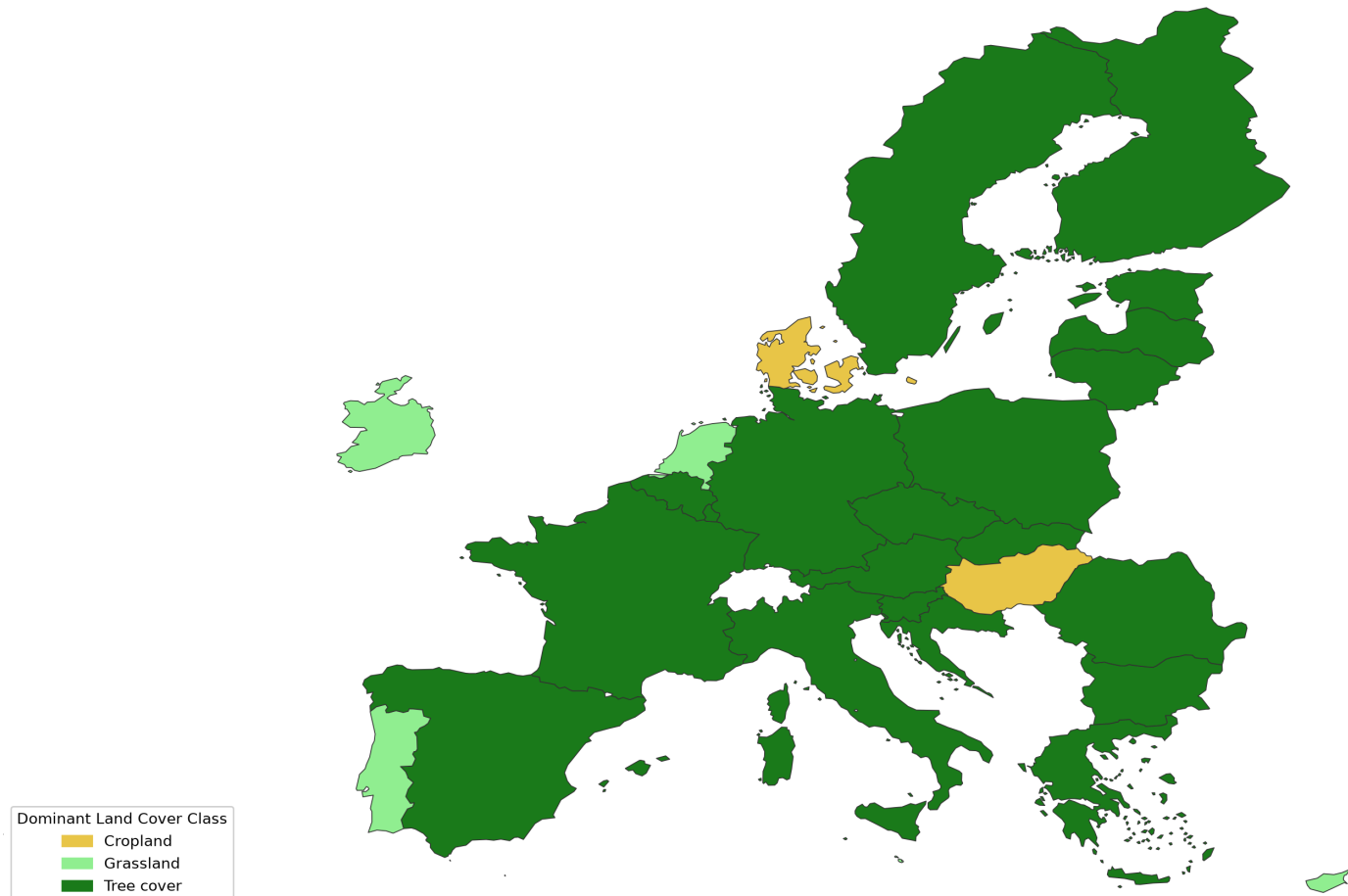
Green Policy Intelligence Engine (GPIE) — Source: Eurostat (EU-27), World Bank (control group)

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Dominant Land Cover Class Map

Control variable - dominant land-cover classification by country.

Dominant Land Cover Class by Country (EU-27)
Each country shaded by its single largest ESA WorldCover class (2021)



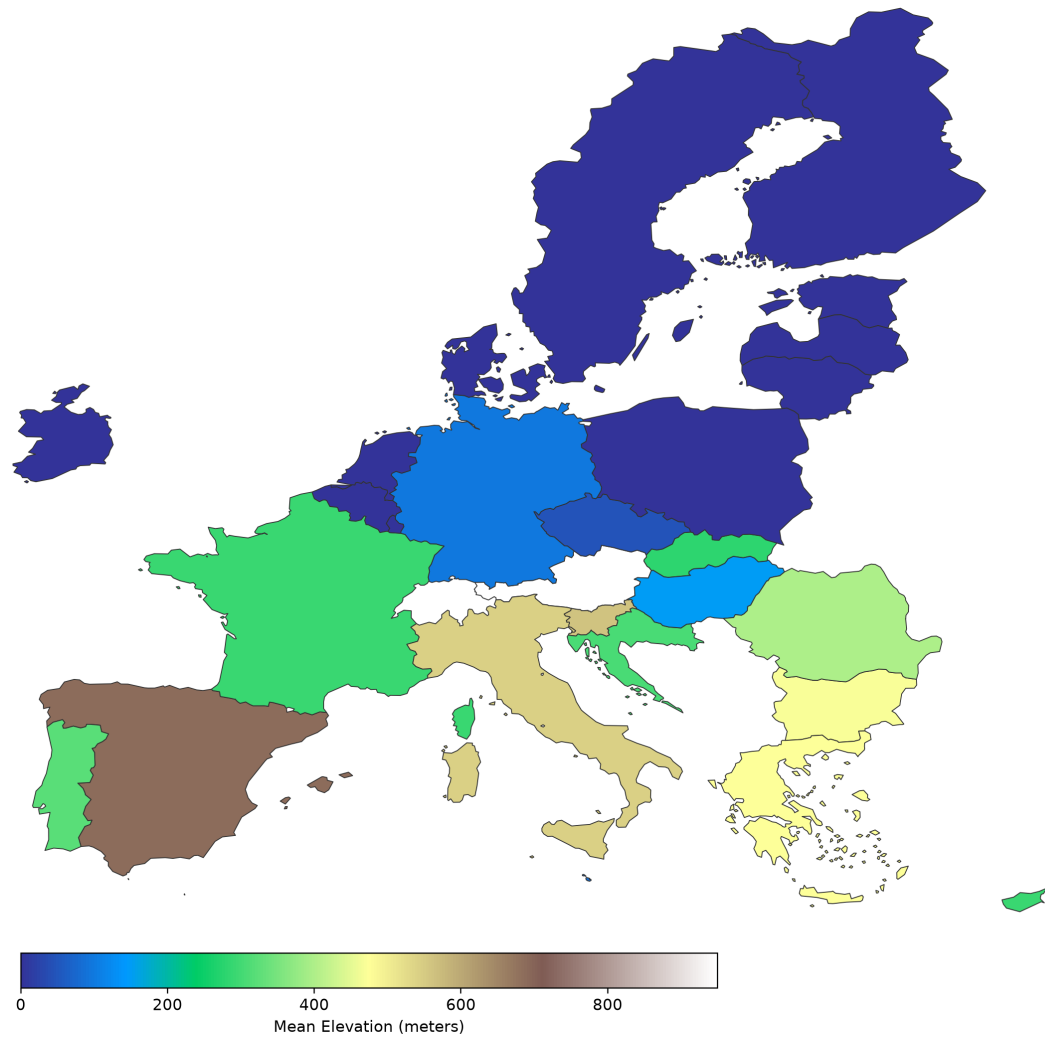
Green Policy Intelligence Engine (GPIE) — Source: ESA WorldCover 10m v200

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Elevation (DEM) Map

Control variable - digital elevation model.

Mean Elevation by Country (EU-27)
Environmental/topographic context — Copernicus DEM GLO-30

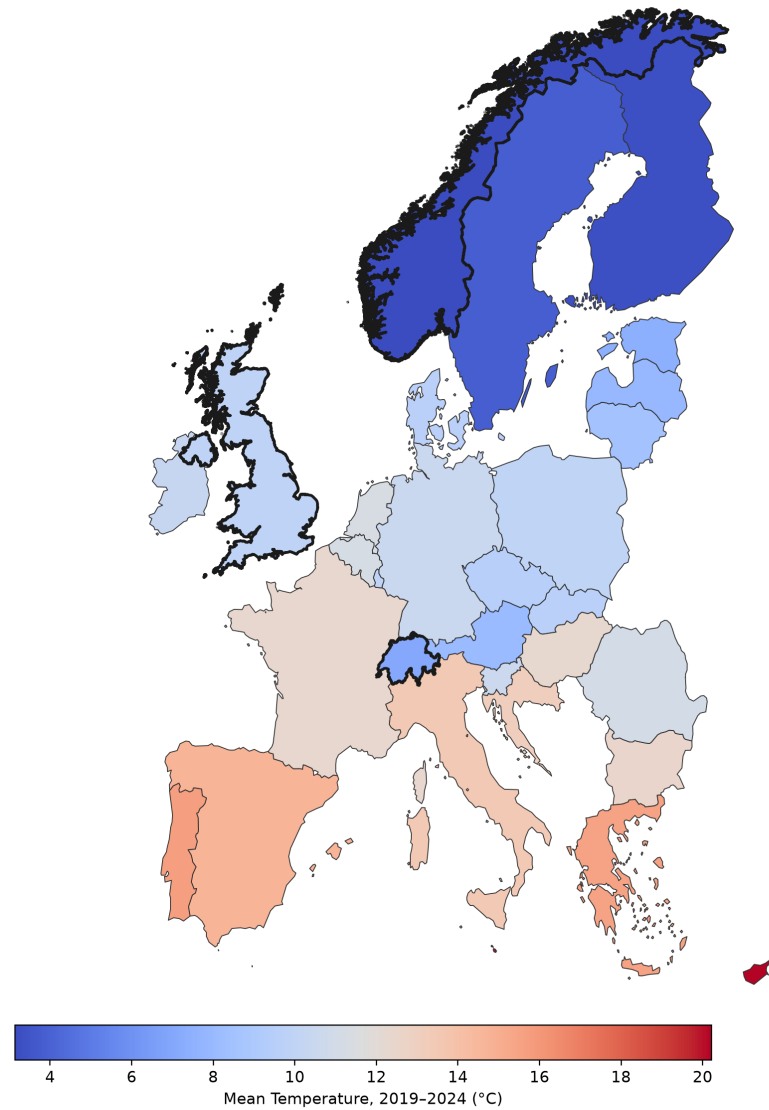


Climate / Temperature Map

Control variable - average temperature by country.

Mean Temperature: EU-27 vs. Control Group (UK, Norway, Switzerland)

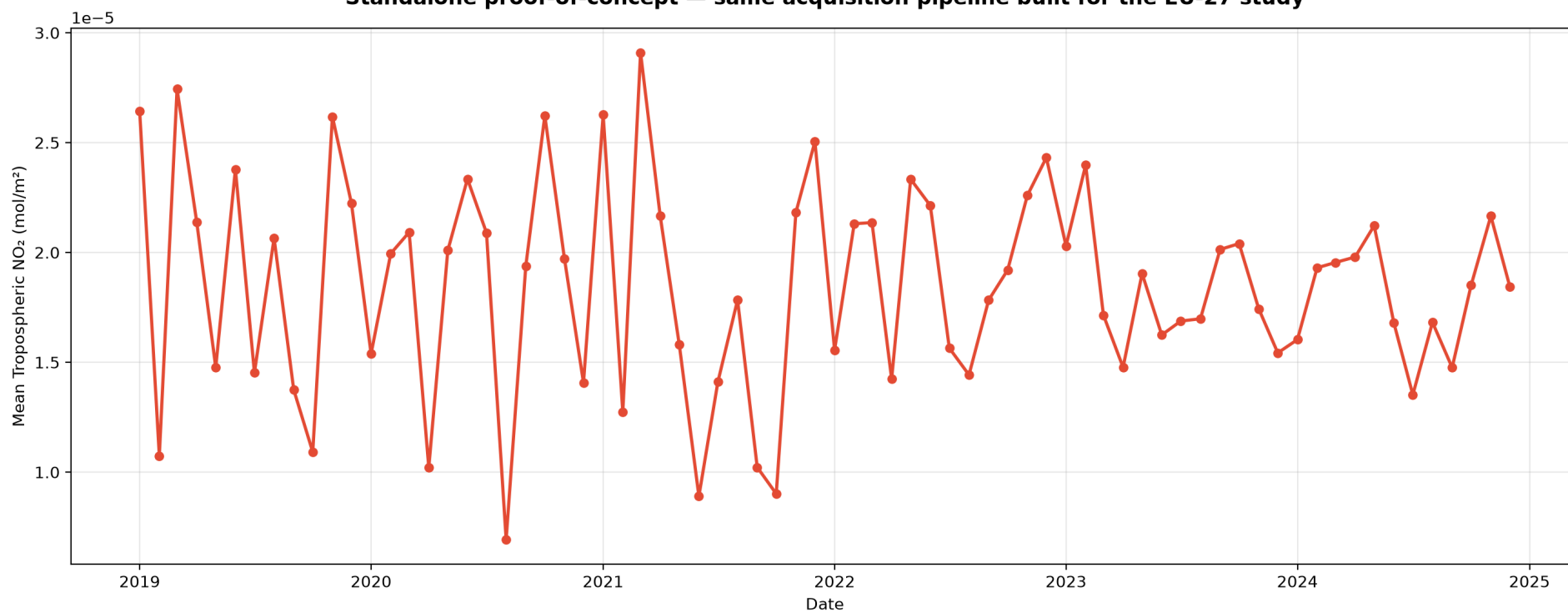
Thick borders mark non-EU control-group countries — temperature used as a climate control variable in the causal model



India Transferability Validation

NO2 acquisition pipeline independently tested on India, 2019-2024.

GPIE Transferability Test: India NO₂ Trend (2019-2024)
Standalone proof-of-concept — same acquisition pipeline built for the EU-27 study

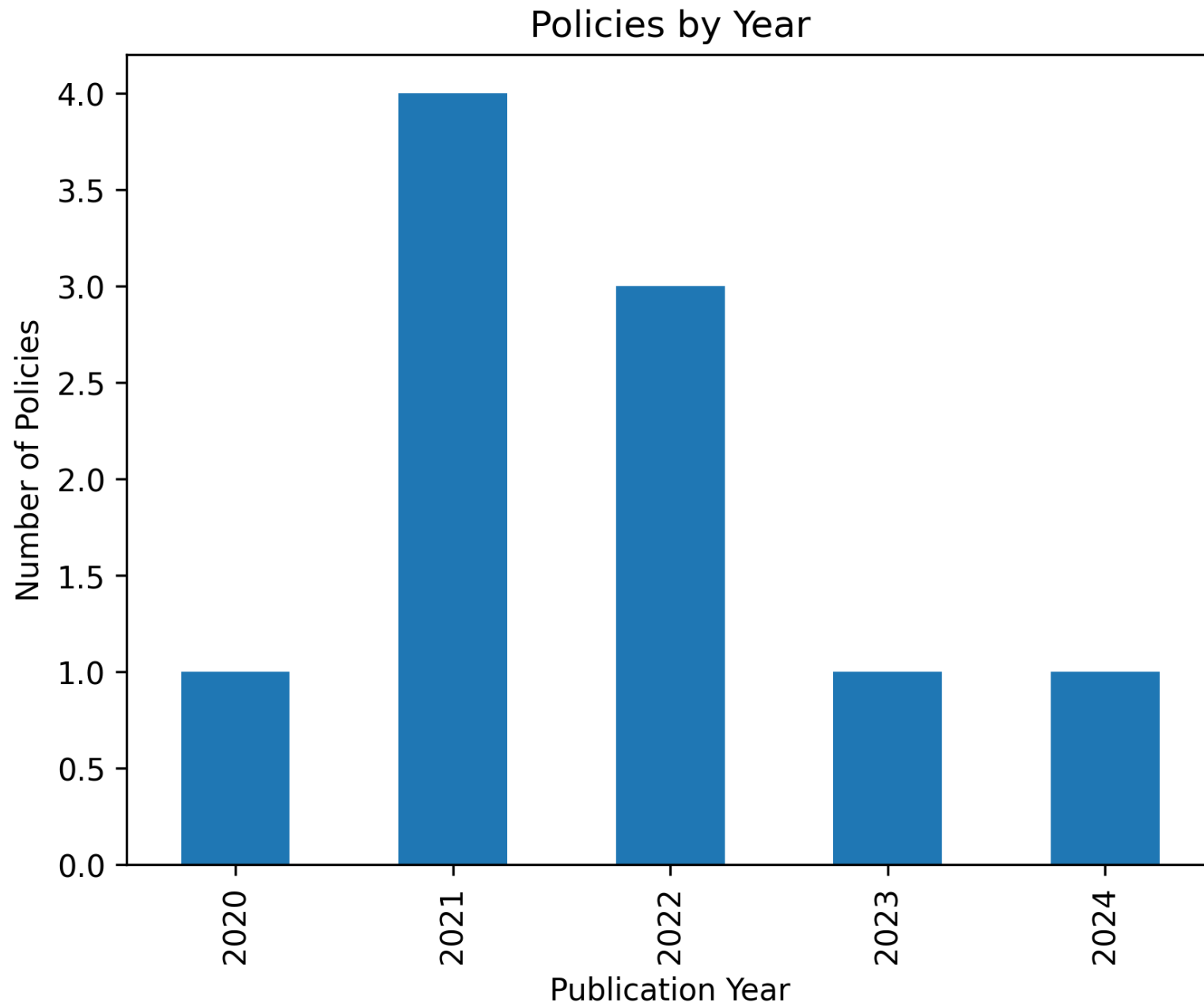


Green Policy Intelligence Engine (GPIE) — Source: Sentinel-5P TROPOMI, Sentinel Hub Statistical API

GPIE - Green Policy Intelligence Engine

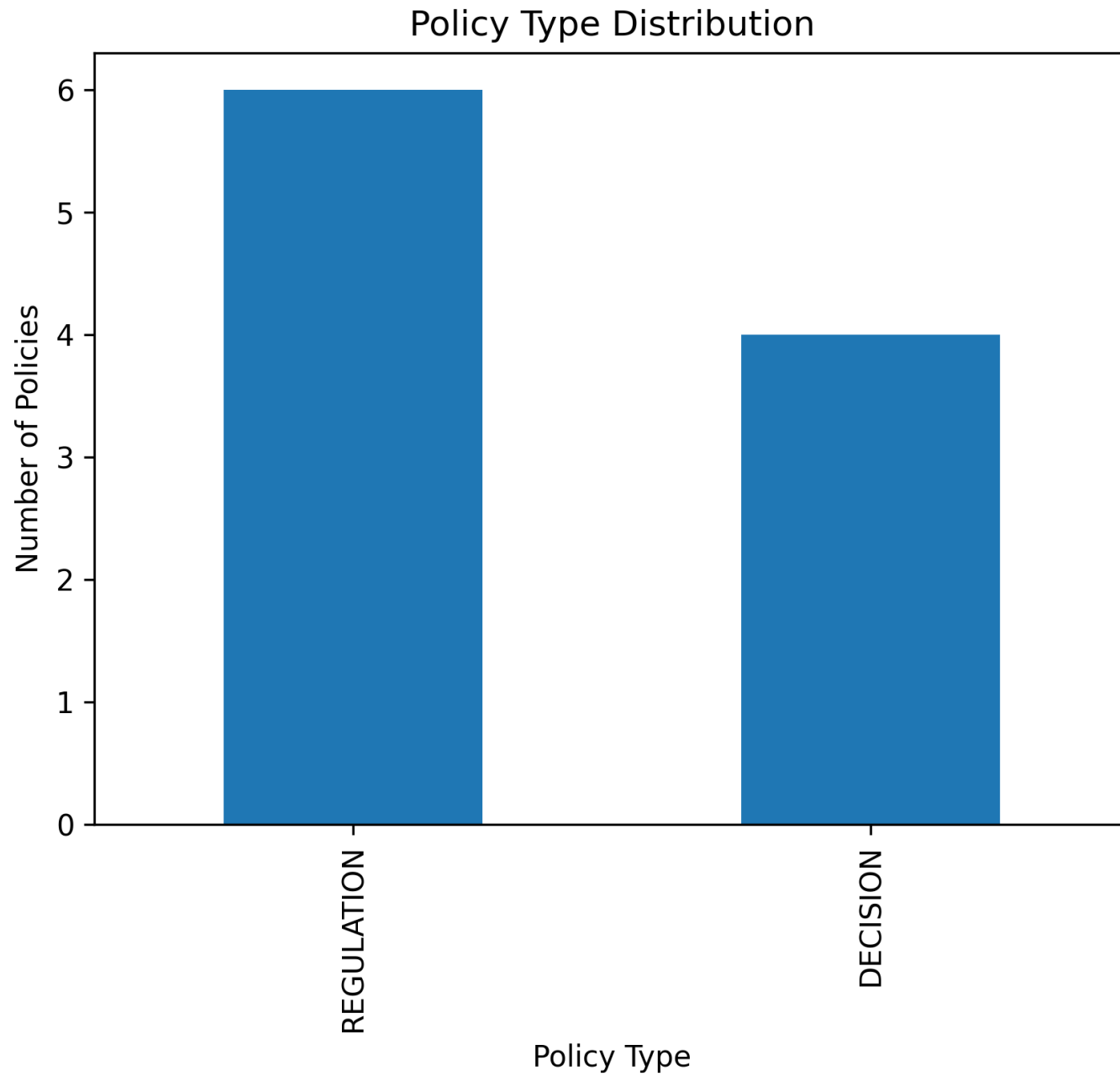
Policies by Year

Count of EU environmental/climate policies enacted per year.



Policy Type Distribution

Breakdown of policy dataset by policy type.



Policy Types by Year

Policy type composition over time.

